INSTRUCTOR COPY **Cryptography All-in-One: TEACHER VERSION**

This is the pupil booklet with the lesson timeline added (this page) and the answers filled in, in **red**. Page numbers match the pupil booklet (page 1 is its cover). Pupils never see this version.

Timeline 90 minutes: what happens when

minutes	activity	slides	resource
0–10	**Talk.** Hook: encoding, you already do it, why compress, what RLE is. Skim 3 to 5, 8 and 10 if short of time.	1–11	slides
10–12	**Worked example.** Decode the first picture together on the board.	12–13	slides
12–22	**Pupils, by hand.** Warm-up rows, mystery 1, mystery 2 (3 and 4 are stretch).	14	booklet pp. 2–5
22–32	**Talk.** Quick answer check, then "that took a while": a row is a list, a picture is a list of rows, `show_picture` and the heart (live in Spyder), decoding one row.	15–16, 21–33	slides, Spyder live
32–37	**Pupils.** Open Spyder, run the starter (Task 1), trace a row on paper, plan in English.	41	booklet pp. 6–8, code
37–42	**Talk.** `.append`, a loop inside a loop, the shape of `decode_row`.	34–37	slides
42–57	**Pupils, code.** Write `decode_row` and test it on `[2, 7, 3]` (Task 2). Hints in layers.	36–37, 41	booklet p. 9, code
57–60	**Talk.** One clue becomes a row, so many clues become a picture: `decode_picture`.	38–40	slides
60–75	**Pupils, code.** Write `decode_picture`, show the first picture (Task 3), then the big pictures (Task 4).	41	booklet pp. 10–11, code
75–80	**Talk.** The finished product: the same code on much bigger pictures. Show what the plots should look like.	44–50	slides
80–88	**Pupils.** Time the big pictures, compare with by hand (Task 5). **Stretch:** `encode_row`, Encode it.	43	booklet pp. 11–12
88–90	**Recap.** What you learned: encoding, compression, RLE, loops and functions.	51–52	slides

Notes For the teacher

- **It is tight.** The code is the point: if the by-hand block overruns, stop it at 22 minutes. Mystery 3 and 4 are for fast finishers, and answers are on slides 17 to 20.
- **Pupils working ahead are fine.** The booklet has every instruction and three layers of hints, so pupils can move on while you talk to others. Do not wait for the slowest pair before each talk: keep talks to the minutes above.
- **Common slips:** swapping the colour inside the inner loop (stripes), a missing `return` (prints `None`), `append` indented in the wrong loop. Answers: `worksheets/teacher/Cryptography-All-in-One-answers.pdf`.
- **Assessment:** watch Task 2 working (`decode_row` on `[2, 7, 3]`) and ask how the swap works. Task 5 times and the written reasons on booklet page 11 show understanding of why computers help.

Cryptography Challenge

Run-length encoding: decode pictures by hand, then teach Python to do it.

Name: _____

Class: _____

Date: _____

Today What we are doing

what we do	booklet
Listen: what is encoding, and why do computers compress data?	
Watch: decode the first picture together	
By hand: shade pictures from their clues	pages 2 to 4 (5 if you finish)
Listen: how could a computer do this for us?	
Spyder: run the starter code, trace a row on paper	pages 6 to 8
Listen: `.append` and a loop inside a loop	
Code: write `decode_row`	page 9
Listen: now a whole picture	
Code: write `decode_picture`, then the big pictures	pages 10 and 11
Listen: the finished product	
Code: how fast is the computer? Stretch tasks	pages 11 and 12
Recap: what you learned today	

How it works Use this booklet your way

- Work at **your own speed**. When your teacher is talking, pens and keyboards down and listen.
- Pages 2 to 5 are **by hand** (pencil). Pages 6 to 12 use **Spyder**.
- **Stuck?** Read Hint 1, try again. Then Hint 2, try again. Then Hint 3. Then ask.
- **Ahead of the class?** Keep going. The stretch tasks are on pages 5 and 12.

My progress ☐ by hand (2 to 4) ☐ Task 1 ☐ Task 2 ☐ Task 3 ☐ Task 4 ☐ Task 5 ☐ stretch

PART 1 BY HAND

about 3 min

Warm-up: rows

Clue = blanks, coloured, blanks, . . . Shade the boxes with a pencil.

A Decode the rows

Example: `2, 7, 3` means 2 blank, 7 coloured, 3 blank.

2, 7, 3



Your turn. (0 first = starts coloured)

3, 5, 4



0, 4, 8



1, 2, 2, 2, 5



6, 6



0, 3, 3, 3, 3



B Squash it

Write each string as a clue: count each run, then the letter.

`AAAABBBCCCCD` → 4A 3B 4C 1D

`WWWWWWBBW` → 6W 2B 1W

`XXXXXXXXXX` → 10X

C Stretch it

`4C 2A 5B` → C C C C A A B B B B B

`3X 1Y 2Z` → X X X Y Z Z

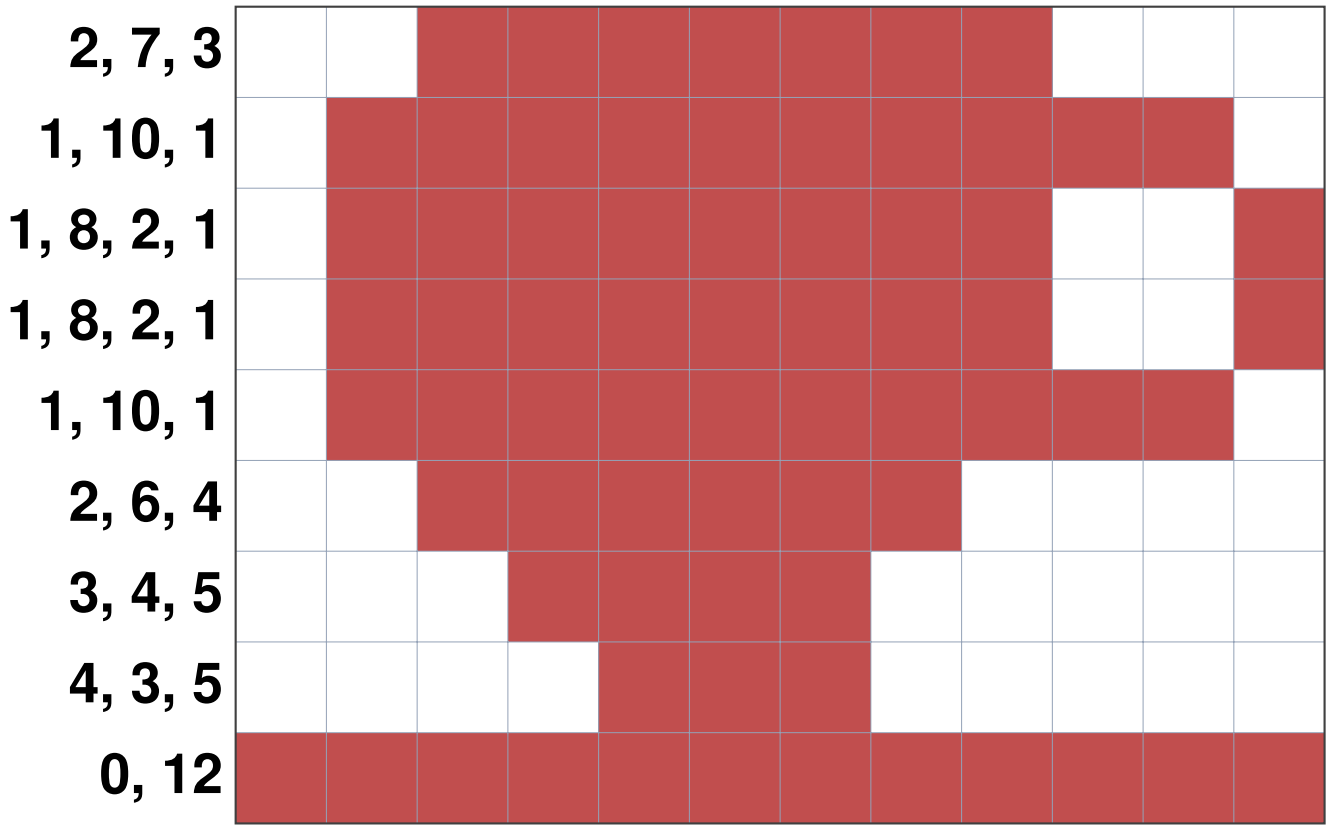
`1A 1B 1A 1B 1A` → A B A B A

PART 1 BY HAND

about 4 min

Mystery picture 1

Shade the coloured pixels. Clue = blanks, coloured, blanks, . . .



Tip: a clue that starts with `0` means the row starts coloured. Every row's numbers add up to 12.

What is it? _____

How long did it take? _____ min

ANSWER Mystery picture 1 is a **mug**. Every row adds up to 12.

PART 1 BY HAND

about 3 min

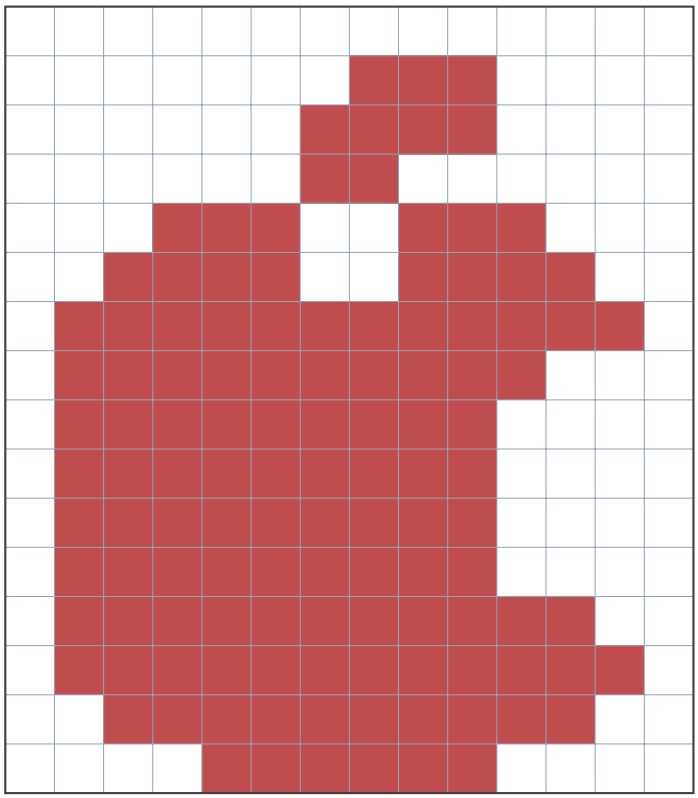
Mystery pictures 2 and 3

Do 2 first. Do 3 if you have time. Each row adds up to 14.

Mystery picture 2

What is it? _____

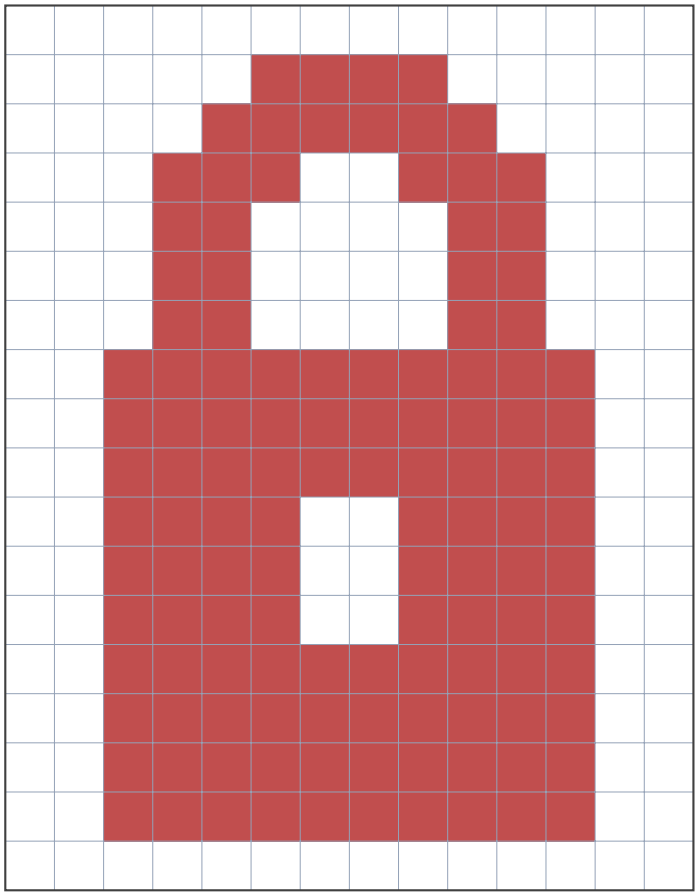
14
7, 3, 4
6, 4, 4
6, 2, 6
3, 3, 2, 3, 3
2, 4, 2, 4, 2
1, 12, 1
1, 10, 3
1, 9, 4
1, 9, 4
1, 9, 4
1, 9, 4
1, 11, 2
1, 12, 1
2, 10, 2
4, 6, 4



Mystery picture 3

What is it? _____

14
5, 4, 5
4, 6, 4
3, 3, 2, 3, 3
3, 2, 4, 2, 3
3, 2, 4, 2, 3
3, 2, 4, 2, 3
2, 10, 2
2, 10, 2
2, 10, 2
2, 4, 2, 4, 2
2, 4, 2, 4, 2
2, 4, 2, 4, 2
2, 10, 2
2, 10, 2
2, 10, 2
2, 10, 2
14

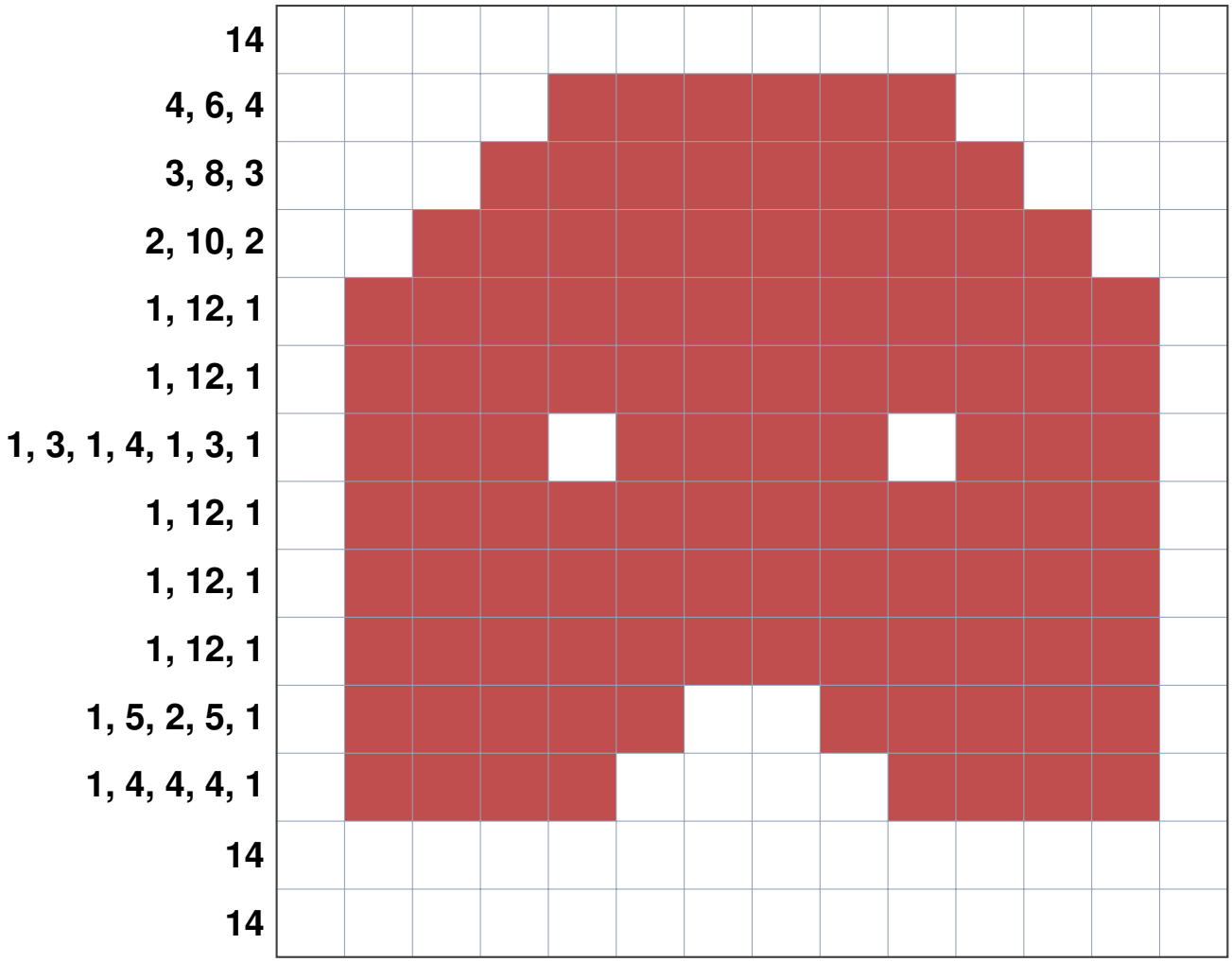


ANSWER Mystery picture 2 is the **Apple logo**. Mystery picture 3 is a **padlock**.

if you finish early

Mystery picture 4

Finished early? Shade this one and answer the pattern questions.



? Spot the pattern

1. Which rows have the same clue? Rows _____ and _____

2. Which row is the widest (most coloured pixels)? Row _____

3. What do you notice about the left and right sides of the picture?

ANSWER Same clue: rows 1 and 13 and 14; rows 5 and 6 and 8 and 9 and 10. Widest row: row 5, 6, 8, 9, 10. Left and right: the picture is a mirror image (every row reads the same backwards).

PART 2 SPYDER

about 2 min

Get Python ready

Do each step, then tick the box.

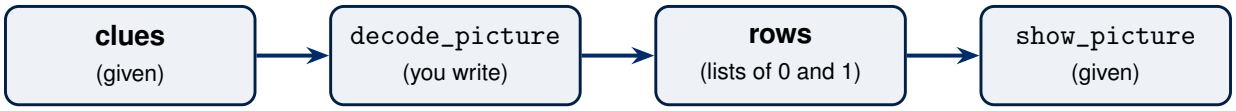
1. **Open Spyder.**
2. **Open the file** `rle_starter.py` . Your teacher will tell you which folder it is in: _____
3. **Check** `pictures.py` is in the *same* folder. You never change that file.
4. **Run it:** press the green triangle (or **F5**).
5. **Look at the Plots pane.** You should see a heart made of dark and blank squares.

☐ **Task 1 done:** I can see the heart.

Check: a heart in the Plots pane. No heart? `pictures.py` **is in a different folder, or the Plots pane is hidden (View, Panes, Plots).**

Remember How the computer sees a picture

- A **row** is a list of 0s and 1s: `row = [0, 0, 1, 1, 1, 0]` . 0 = blank, 1 = coloured.
- A **picture** is a list of rows.
- A **clue** is a list of numbers: blanks, coloured, blanks, . . .
  `[2, 7, 3]` becomes `[0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0]` .



Tasks Your five tasks for today

task	what to do	page	done
1	`rle_starter.py` : run it, see the heart	6	☐
2	finish `decode_row` , test it on `[2, 7, 3]`	9	☐
3	finish `decode_picture` , show the picture from Mystery 1	10	☐
4	`DINOSAUR` , `MARIO` , then `UK_IRELAND`	11	☐
5	`UK_IRELAND` : how long by hand? How long by computer?	11	☐

PART 2 PAPER FIRST

about 2 min

Be the computer

Follow the code one step at a time. Fill in every empty box.

A Trace `decode_row([2, 7, 3])`

Start: `row = []` `colour = 0`. Step 1 is done for you.

step	number	colour	what gets added	row so far
1	2	0	`0, 0`	`[0, 0]`
2	7	**1**	**1 seven times**	`[0, 0, 1, 1, 1, 1, 1, 1, 1]`
3	3	**0**	**0 three times**	`[0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0]`

B A clue that starts with 0

Trace `decode_row([0, 4, 2])`. Fill in the rest.

step	number	colour	what gets added	row so far
1	0	0	**nothing**	`[]`
2	4	**1**	**1 four times**	`[1, 1, 1, 1]`
3	2	**0**	**0 twice**	`[1, 1, 1, 1, 0, 0]`

? What is different about the first step?

`range(0)` **runs zero times, so nothing is added, but the colour still swaps. The row starts with 1.**

What does `colour = 1 - colour` **do?** Try `colour = 0`, then `colour = 1`.

It flips the colour: 0 becomes 1 and 1 becomes 0 ($1 - 0 = 1$ and $1 - 1 = 0$).

PART 2 PLAN

about 1 min

Plan before you type

Write the steps in plain English first. No code yet!

A Steps for `decode_row`

Write the steps the function follows, one per line.

Word bank: start • for each number • add • swap colour • return

1. **Start with an empty row and colour 0.**
2. **For each number in the clue:**
3. **add that many copies of colour to the row**
4. **swap colour (0 to 1, 1 to 0)**
5. **return the row**

B Steps for `decode_picture`

Same again, for a whole picture.

1. **Start with an empty picture.**
2. **For each clue: decode it into a row**
3. **add the row to the picture**
4. **return the picture**

Compare with your trace on page 7: does your plan match what you did by hand?

PART 3 CODE

about 15 min

Task 2: decode_row

Finish the function, run the file, check the test.

```
def decode_row(clue):
    row = []        # the row we are building. It starts empty.
    colour = 0      # we always start with blanks (0)

    # go through each number in the clue (a for loop over the list)
    for number in clue:
```

Model answer

```
    for number in clue:
        for i in range(number):
            row.append(colour)
        colour = 1 - colour
```

```
    return row      # what do we give back?
```

`print(decode_row([2, 7, 3]))` expected: `[0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0]`

What did YOU get? _____ ☐ it matches

Hint 1: the idea

You need a loop inside a loop. The outer loop picks each number from the clue. The inner loop repeats that many times.

Hint 2: the pieces

The inner loop is `for i in range(number):` and each time round it does `row.append(colour)`. After the inner loop finishes, swap the colour: `colour = 1 - colour` (or an if / else).

Hint 3: the shape (fill in the gaps)

```
for number in clue:
    for i in range(number):
        row.append(colour)      # add colour to the row
    colour = 1 - colour          # swap the colour (level with the inner for)
```

Nothing printed, or `None`? Check `return row` is at the very end, level with the first `for`.

☐ **Task 2 done:** my `decode_row` gives the right answer for `[2, 7, 3]`.

PART 3 CODE

about 8 min

Task 3: decode_picture

Reuse your row decoder to decode a whole picture.

```
def decode_picture(clues):
    picture = []   # the picture we are building. It starts empty.

    # go through each clue in clues:
    #   1. decode that clue into a row (use your function!)
    #   2. add the row to the picture   hint: picture.append(...)

    return picture
```

Model answer

```
    for clue in clues:
        row = decode_row(clue)
        picture.append(row)
```

Test it: `show_picture(decode_picture(MUG))` (take the `#` off the line in the starter file)

Does the picture in the Plots pane match your Mystery picture 1? ☐ yes ☐ not yet

Hint 1: the idea

A picture is many rows. You already have a function that turns one clue into one row. Use it once for every clue.

Hint 2: the pieces

Loop over `clues`. Inside the loop: `row = decode_row(clue)`, then `picture.append(row)`.

Hint 3: the shape (fill in the gaps)

```
for clue in clues:
    row = decode_row(clue)
    picture.append(row)
```

Empty picture? Check `return picture` is at the very end, level with the first `for`.

Debugging checklist

- ☐ Did I put `return` at the end, indented once?
- ☐ Is my colour swap INSIDE the outer loop but AFTER the inner loop?
- ☐ Is my `append` indented inside the loop it belongs to?
- ☐ Did I run the file again (F5) after changing it?

☐ **Task 3 done:** my code draws the picture from Mystery 1.

about 15 min

Tasks 4 and 5: the finished product

Same code. Much bigger pictures. Then time it.

Task 4 Run your code on the big ones

Take the `#` off these lines in the starter file, one at a time, and run it. Tick each when it works.

```
show_picture(decode_picture(DINOSAUR))      # [ ] worked
show_picture(decode_picture(MARIO))         # [ ] worked
show_picture(decode_picture(UK_IRELAND))    # [ ] worked (the one that took ages by hand!)
```

Did your code have to change? ☐ yes ☐ no **(no: that is the point)**

Task 5 How long does each way take?

picture	rows	did it work?	time by computer	time by hand
Mystery 1	9			min (page 3)
`DINOSAUR`	47	**yes**	**about 1 s**	
`MARIO`	32	**yes**	**about 1 s**	
`UK_IRELAND`	46	**yes**	**about 1 s**	my guess: min

Estimate it. Mystery 1 took me _____ min. `UK_IRELAND` has about 5 times as many rows (46 compared with 9), and its rows are longer. So by hand it would take me at least _____ × 5 = _____ min.

Which is faster for the biggest picture, you or Python? Why?
Python, by far (about a second against many minutes by hand). The same loops do the same job however many rows there are, and the computer never gets tired or loses its place.

Think: what if the picture was a photo with a million rows? What would you do?
Let the computer do it: write the decoding once and the loop repeats it a million times. By hand it would be impossible.

Want to race the computer? Ask your teacher for a big grid to shade by hand while your code runs.

☐ **Task 4 done** ☐ **Task 5 done**

PART 4 STRETCH

stretch

Go backwards: encode_row

Row in, clue out. For when your decoder works and you want more.

```
def encode_row(row):
    clue = []
    # count how many pixels in a row are the same, and add each count to clue.
    return clue

print(encode_row([0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0]))   # should print [2, 7, 3]
```

Model answer

```
colour = 0
count = 0
for pixel in row:
    if pixel == colour:
        count += 1
    else:
        clue.append(count)
        colour = pixel
        count = 1
clue.append(count)
```

Hint 1: the idea

Start by counting blanks (0). Keep two variables: `colour` (starts at 0) and `count` (starts at 0). Loop over every pixel in the row.

Hint 2: the pieces

If the pixel is the same as `colour`, do `count += 1`. Otherwise: add `count` to `clue`, change `colour` to the new pixel, and start counting again at 1.

Hint 3: the easy-to-forget bit

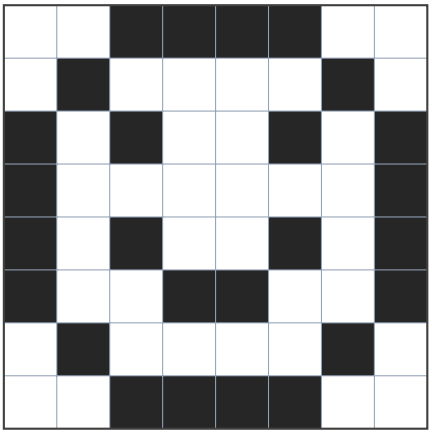
When the loop ends, the last run of pixels has not been added to `clue` yet. Add it before you `return`.

By hand Encode it

Write each row's clue in its box. Blanks first, then coloured (start with `0` if the row starts coloured).

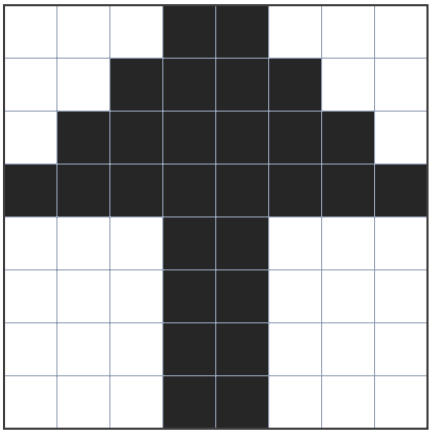
Picture A

2, 4, 2	
1, 1, 4, 1, 1	
0, 1, 1, 1, 2, 1, 1, 1	
0, 1, 6, 1	
0, 1, 1, 1, 2, 1, 1, 1	
0, 1, 2, 2, 2, 1	
1, 1, 4, 1, 1	
2, 4, 2	



Picture B

3, 2, 3	
2, 4, 2	
1, 6, 1	
0, 8	
3, 2, 3	
3, 2, 3	
3, 2, 3	
3, 2, 3	



Think: would clues save more space on a picture with lots of single dots, or with big solid blocks?

Solid blocks: single dots give runs of 1, so the clue is no shorter.

Keep going: design your own picture and clues, and swap with a partner. Ask your teacher for the Design your own sheet.